Ex:No:1
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1. Various Network commands in Linux & Windows.

Aim:

To study of various network commands used in Linux & windows.

Basic NetworKing commands:

1) arp -a : (terminal) ARP is short form of address resolution protocol, it will show the IP address of your computer along with the IP address and MAC address of your router.

Output:

gateway (10.0.2.2) at 52:54:00:12:35:02 [ether] on enp0s3.

2) hostname : (terminal) This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output:

fedora.

3) ipconfig/all : (command)

This command displays detailed configuration information about your TCP/IP connection including router, gateway, DNS, DHCP and type of ethernet adapter in your system.

Output: Hostname: laptop-MHUMQBR6.
Physical address: 0A-00-27-00-00-0A.
DHCP enabled : No
Subnet mask : 255.255.255.0
IPV4 address : 192.168.56.1

(Command).

4) nbtstat -a : This command help solve problems with NetBios name resolution.

Output:

-a : Lists the remote machine's name table given its name.

-n : List local NetBios names;

-r : List names resolved by broadcast and via WINS.

5) netstat : netstat displays a variety of statistics about a computers active TCP/IP Connections.
(Command)

Output:

Proto	Local address	foreign address	state
TCP	10.106.210.192:60268	a23-55-47-124: http.	time_wait.
TCP	10.106.210.192:63441	202:4070	Established.
TCP	127.0.0.1:7768	Laptop-MHUMQBR6: 52692	Established.

6) nslookup : (name server lookup) is a tool used to perform DNS lookups in Linux.
(Command)

Eg: -nslookup: www.google.com.

Output:

Default Server : Unknown.

Address : 10.106.210.238.

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7) pathping : Pathping is unique to window's and is basically a combination of ping and tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way, gathering statistics on the rate of data loss along each hop.

Output:

Server : 10.106.210.238.

Address : 10.106.210.238#53.

8) ping : (Packet INternet groper) command is the best way to test connectivity between two nodes. Ping use ICMP (Internet Control Message Protocol) to communicate to each other devices.

Output:

64 bytes from fedora (fe 80::96 ff:454d:ce50) : icmp-seq =
ttl=64 time = 0.088 ms.

PING 24.21.4.8 (24.21.4.8) 56(84) bytes of data.

9) Route : Command is used to show / manipulate the IP routing table. It is primarily used to setup static routes to specific host or networks via an interface.

Output:

Kernel IP routing table

Destination	Gateway	Genmask	Flags	Metric	Ref	Use
default	-gateway	0.0.0.0	UG	100	0	0
10.0.2.0	0.0.0.0	255.255.255.0	U	100	0	0

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Some important Linux networking commands:

1) ip.

* Every administrator will need in daily work, from setting up new systems and assigning IPs to troubleshooting existing systems.

Eg: ip address shown

link (lookup 00:00:00:00:00:00 and
00:00:00:00:00:00)
inet (127.0.0.1.18' scope host lo).

2) if config:

It is a staple in many sysadmin's tool belt for configuring and troubleshooting networks, It has since been replaced by the ip command discussed above.

3) mtr:

* Serves as a network diagnostic and trouble shooting tool.

Eg: mtr google.com

Packets:

loss	snt	last	avg	best	Wst
0.0%	178	0.1	0.1	0.0	0.1

4) tcp dump.

Designed for capturing and displaying packets.

Eg: tcpdump -D

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5) ping.

* It is a tool that verifies IP-level connectivity to another TCP/IP computers by sending ICMP echo request messages.

Eg: ping google.com,

Student Observation:

1) Which command is used to find the reachability of a host machine from your device?

⇒ ping command is used to find the reachability of a host machine from your device.

2) Which command will we give the details of hops taken by a packet to reach its destination?

⇒ mtr, it shows route from a computer to a specified host mtr provides a lot of statics about each hop such as response time and percentage.

3) Which commands displays the ip configuration of your machine.

⇒ ip command can show address information manipulate routing plus display network various devices interfaces & funnels.

4) Which command displays the TCP port states in your machine?

⇒ net stat.

It displays variety of statistic about a computer active TCP/IP connection.

5) Write the modify the ip configuration in a Linux machine.

⇒ * Assigning IP address to interfaces ip address add 192.168.1.254/24.

* Deleting IP address.

ip address del 192.168.1.254/24. device enp-S03.